

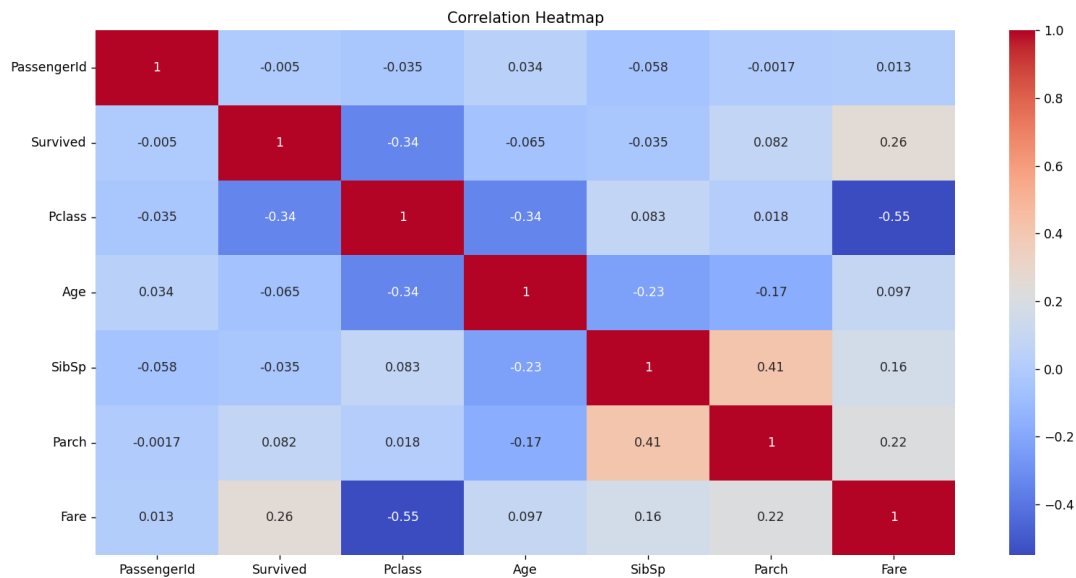
Report of Findings

Tools: Python (Pandas, Matplotlib, Seaborn)

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age         714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

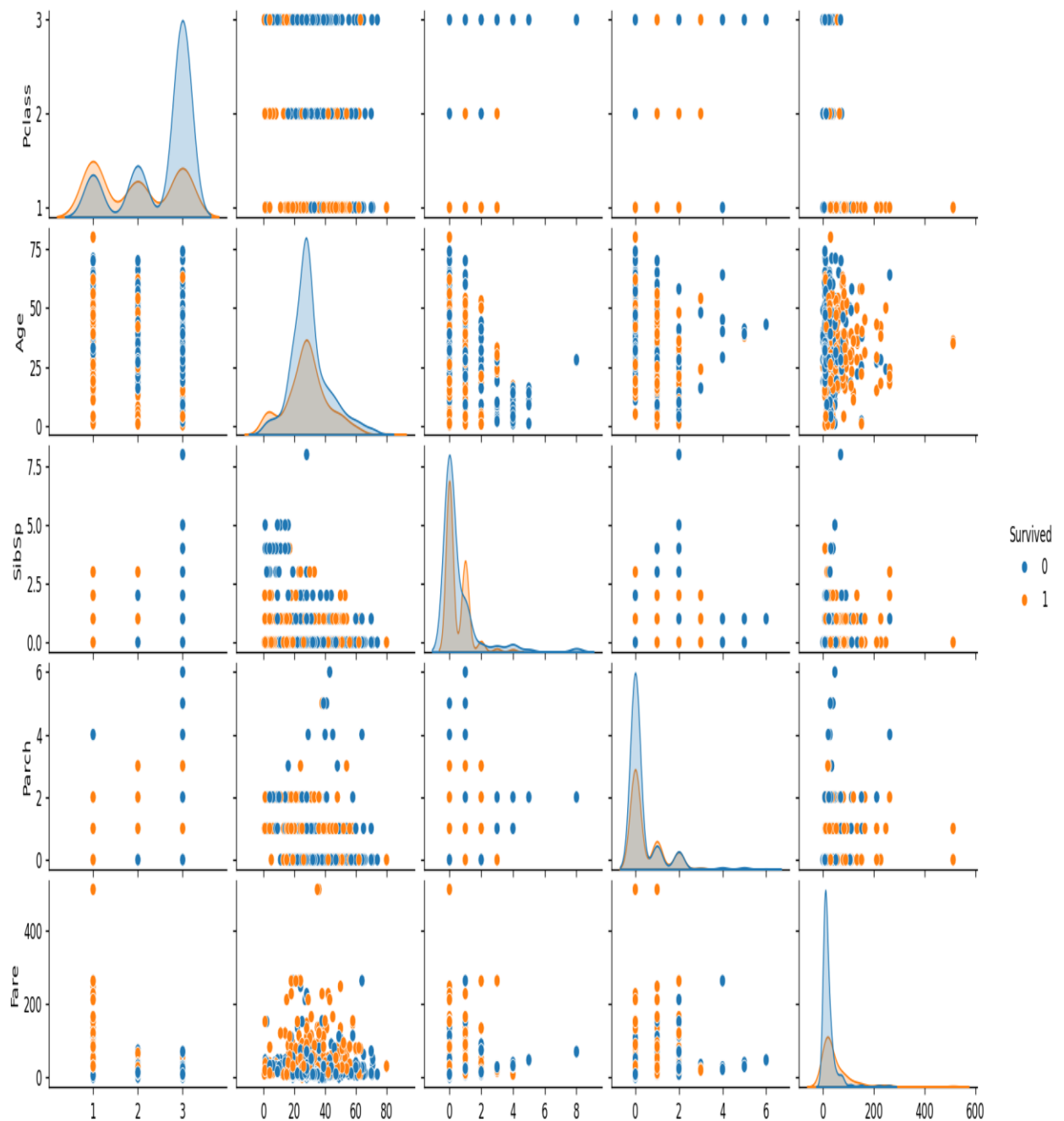
	PassengerId	Survived	Pclass	...	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	...	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	...	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	...	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	...	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	...	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	...	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	...	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	...	8.000000	6.000000	512.329200

```
[8 rows x 7 columns]
Survived
0    549
1    342
Name: count, dtype: int64
Pclass
3     491
1     216
2     184
Name: count, dtype: int64
Sex
male      577
female    314
Name: count, dtype: int64
```

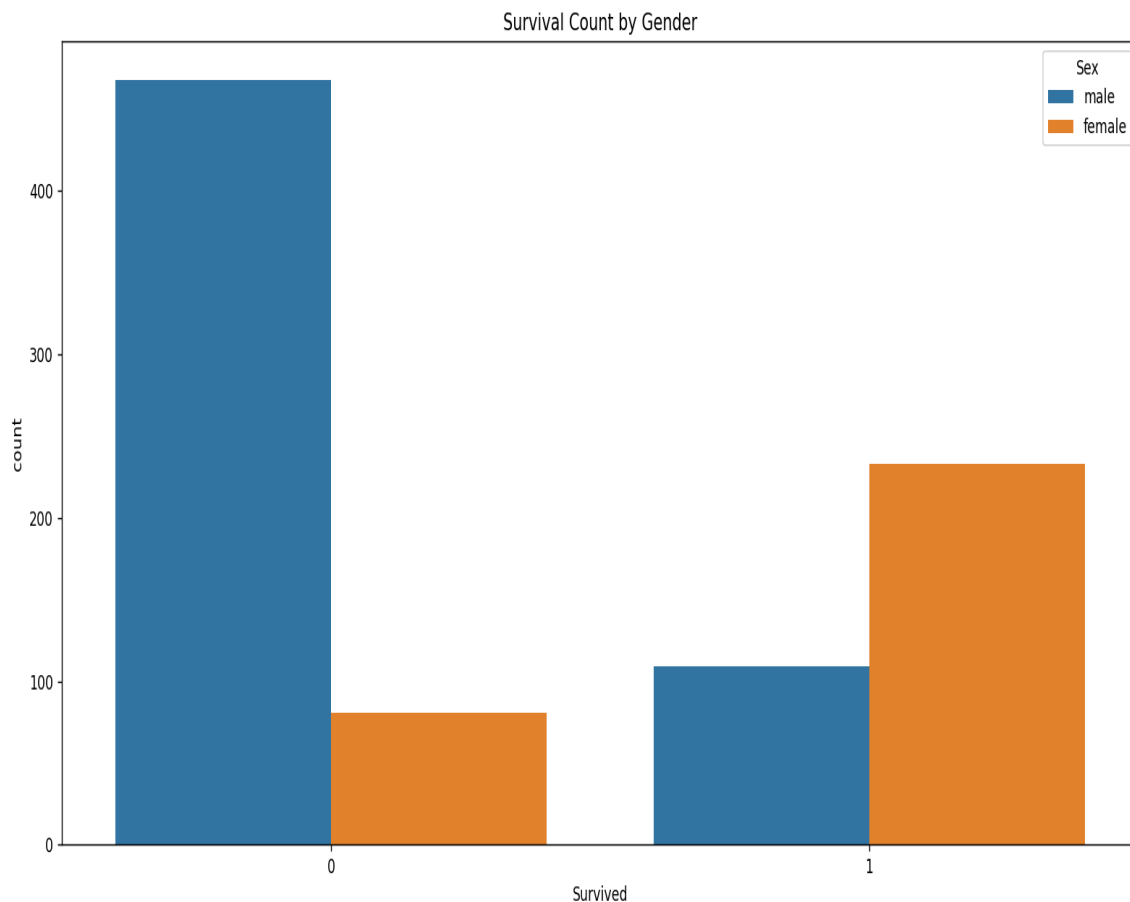


This correlation heatmap visually represents the linear relationships between different features in a dataset, likely related to the Titanic passenger survival. Key takeaways include:

- * Passenger class and fare seem to be important factors influencing survival. Higher class and higher fare were associated with a greater likelihood of surviving.
- * There's a logical connection between passenger class and fare, with higher classes paying more.
- * The number of siblings/spouses aboard tends to be related to the number of parents/children aboard.
- * Other factors like age and the number of siblings/spouses or parents/children alone don't show a strong linear relationship with survival.



This pair plot visualization of the Titanic dataset strongly suggests that several passenger characteristics were associated with the likelihood of survival. Higher passenger class and a higher fare paid appear to be significant positive indicators of survival. Additionally, having fewer family members aboard (in terms of siblings/spouses and parents/children) also seems to correlate with a higher survival rate. While the relationship between age and survival is less definitive, there's a subtle indication that younger passengers might have had a slightly better chance. These observations align with common narratives about the Titanic disaster, where factors like social status and the ability to access resources (indicated by fare) played a crucial role in survival.



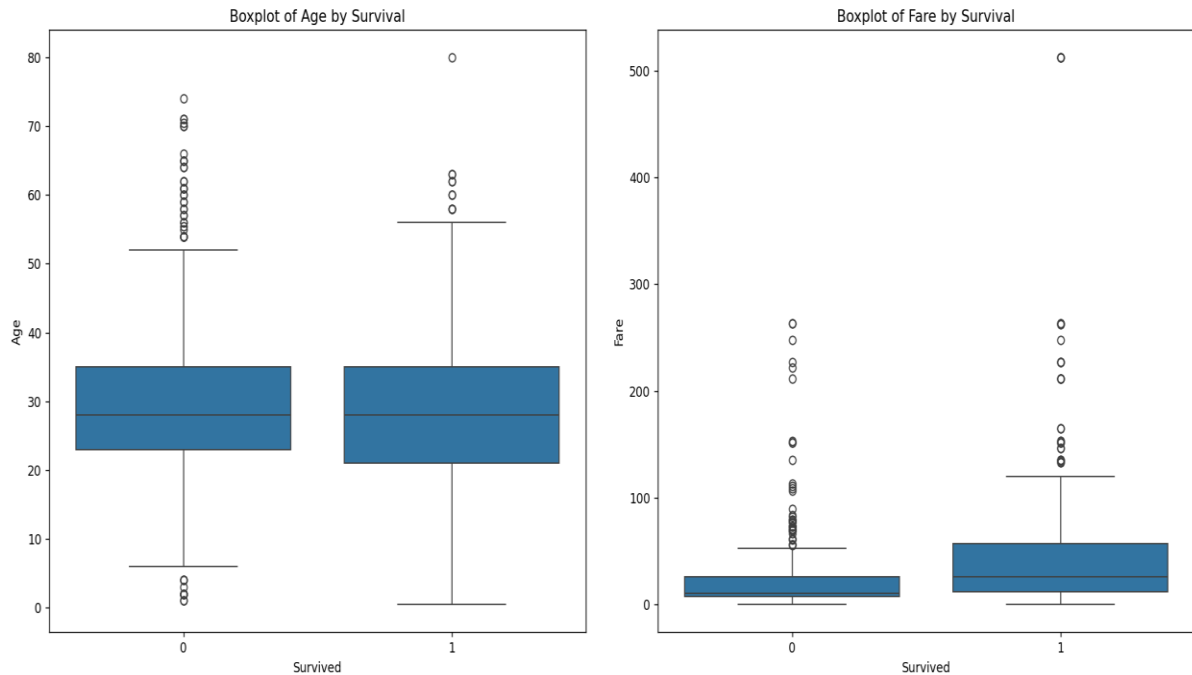
* The chart displays the count of passengers based on their survival status (0 = Did Not Survive, 1 = Survived) and gender (blue for male, orange for female).

* For those who did not survive (Survived = 0), the number of males (blue bar) is significantly higher than the number of females (orange bar). The male bar is much taller.

* For those who survived (Survived = 1), the number of females (orange bar) is substantially higher than the number of males (blue bar). The orange bar is taller in this group.

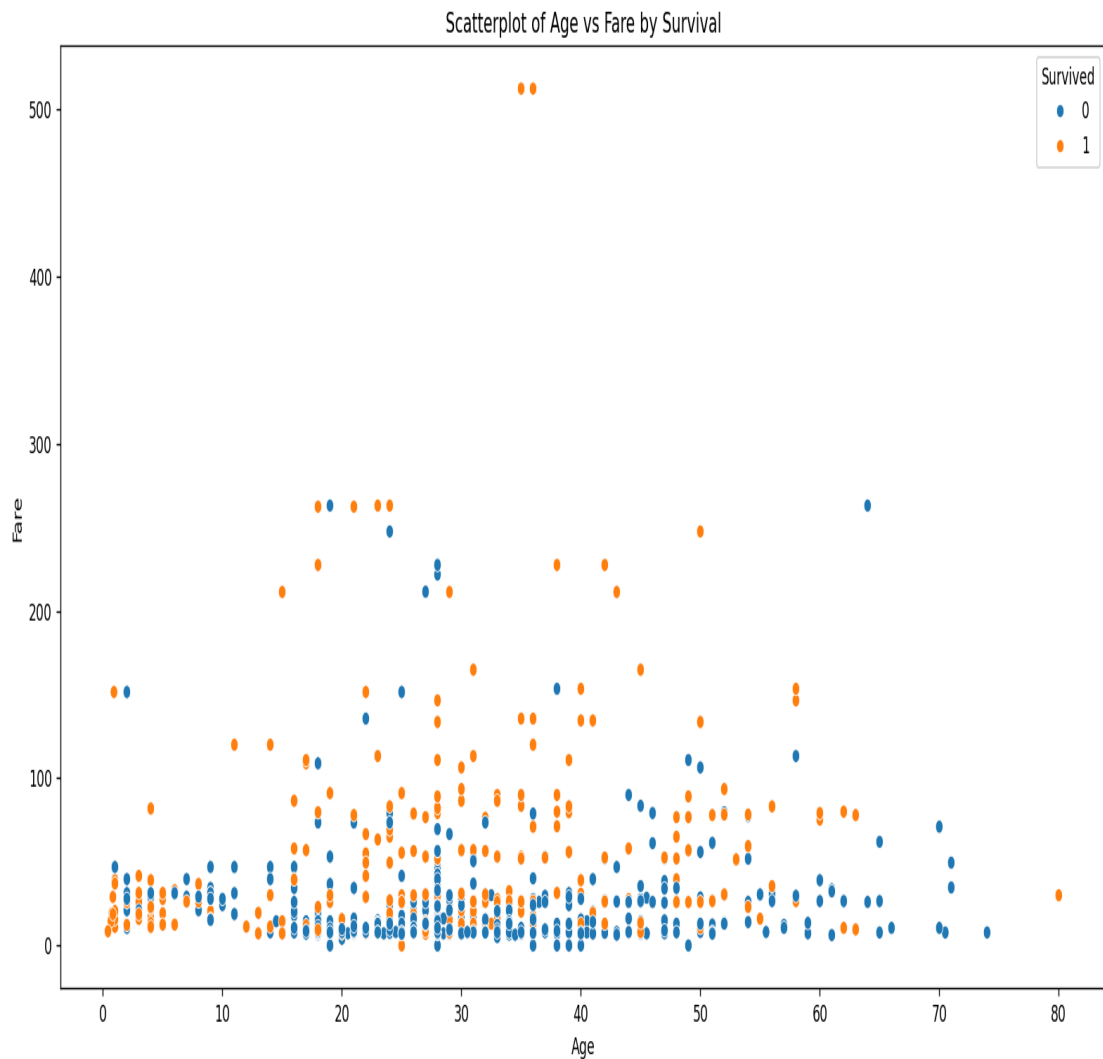
* The total number of males in the dataset appears to be larger than the total number of females, given the height of the male bar in the 'Did Not Survive' category.

* The survival rate seems to be much higher for females compared to males, as indicated by the larger female survivor count relative to the female non-survivor count, and the opposite trend for males.



* **Age and Survival:** The age distribution of survivors and non-survivors is quite similar, with comparable median ages and spread. While there are very young and old individuals in both groups, age doesn't appear to be a primary distinguishing factor for survival based on these box plots alone.

* **Fare and Survival:** There is a clear and substantial difference in the fare paid between the two groups. Passengers who survived generally paid significantly higher fares than those who did not. This reinforces the earlier observation from the pair plot, suggesting that fare (likely correlated with passenger class and access to better accommodations) played a crucial role in survival. The wider range of fares among survivors indicates that survival wasn't limited to just the absolute highest fare-paying passengers, but a generally higher fare was a positive indicator.



This scatterplot visualizes the relationship between the age and fare paid by Titanic passengers and their survival status. The plot suggests that the fare paid might have played a more significant role in survival than age. While passengers of all ages experienced both survival and non-survival, those who paid higher fares appear to have had a greater chance of survival. Specifically, the cluster of passengers who paid the lowest fares shows a mix of both survivors and non-survivors, whereas at higher fares, the proportion of survivors seems to increase. The few passengers who paid the very highest fares all survived in this dataset. Age, on the other hand, doesn't show a clear direct correlation with survival based on this visualization alone, as survivors are distributed across various age groups.